



**उत्पाद मैनुअल
निमज्जन पम्पसेटों के लिए लाइन**

**संचालित ए.सी. मोटरें- विशिष्टि (तीसरा पुनरीक्षण)
आई एस 9283 : 2024 के अनुसार**

**PRODUCT MANUAL
LINE OPERATED A.C. MOTORS FOR
SUBMERSIBLE PUMPSETS —
SPECIFICATION (THIRD REVISION)
ACCORDING TO IS 9283: 2024**

विभिन्न उत्पादों के लिए भारतीय मानक ब्यूरो (अनुरूपता मूल्यांकन) विनियम, 2018 की योजना- I के तहत प्रमाणन के संचालन में एकरूपता और पारदर्शिता के लिए इस उत्पाद मैनुअल का उपयोग सभी क्षेत्रीय / शाखा कार्यालयों और लाइसेंसधारियों द्वारा संदर्भ सामग्री के रूप में किया जाएगा। दस्तावेज़ का उपयोग बीआईएस प्रमाणन प्राप्त करने के इच्छुक संभावित आवेदकों द्वारा भी किया जा सकता है।

This Product Manual shall be used as reference material by all Regional/Branch Offices & licensees to ensure uniformity of practice and transparency in operation of certification under Scheme-I of Bureau of Indian Standards (Conformity Assessment) Regulations, 2018 for various products. The document may also be used by prospective applicants desirous of obtaining BIS certification.

1.	मानक संख्या IS No.	:	IS 9283 : 2024
	शीर्षक Title	:	Line operated a.c. Motors for Submersible Pumpsets —Specification (Third Revision)
	संशोधनों की संख्या No. of amendments	:	Nil
2.	नमूना दिशानिर्देश Sampling Guidelines:		
a)	कच्चा माल Raw material	:	Table 1 of IS 9283:2024
b)	समूहीकरण दिशानिर्देश Grouping Guidelines	:	Please refer Annex – A
c)	नमूने का परिमाण Sample Quantity	:	1 number

3.	परीक्षण उपकरणों की सूची List of Test Equipment	:	Please refer Annex – B
4.	निरीक्षण और परीक्षण की स्कीम Scheme of Inspection and Testing	:	Please refer Annex – C
5.	एक दिन में संभावित परीक्षण Possible tests in a day:	:	All tests as per IS 9283:2024
6.	लाइसेंस का दायरा /Scope of the Licence	:	Licence is granted to use Standard Mark as per IS 9283 : 2024 with the following scope:
	Name of the Product		Line Operated a.c. Motors for Submersible Pumpsets
	Model No.		_____
	Rating		_____kW, ___V, _____Hz, ___Phase
	Bore size		_____mm
	No. of poles		2/4
	Efficiency class		EL1/EL2/ EL3/ EL4/ EL5

ANNEX – A
Grouping Guidelines

1. Manufacturer shall submit the declaration for each of the parameters as specified in the scope of the licence (as per Sl. No. 6 above) for each model of 'the motor' that the manufacturer intends to cover in the scope of licence.
2. For considering GoL/CSoL each model of the motor has to be tested.
3. The Firm shall declare the varieties of motors they intend to cover in the Licence. The Scope of Licence may be restricted based on the Manufacturing capability and Testing facilities of the Manufacturer.
4. During the operation of the Licence, BO shall ensure that all the models covered in the Licence are tested in rotation to the extent possible.
5. Along with the declarations at Sl. 1 above, manufacturer shall also declare the following for each of the model(s):
 - a. Category of motor - A/B
 - b. Types of bearings- anti friction/journal
 - c. Type of motors and declared full load torque

ANNEX – B**परीक्षण उपकरणों की सूची****List Of Test Equipment****(INDICATIVE LIST, FOR GUIDANCE ONLY)*****Major test equipment required to test as per the Indian Standard***

Sl. No	Test Equipment	Tests Used in with Clause reference
1.	Surface Comparator	Cl. 5.5
2.	Dynamic Balancing Machine	Cl. 5.6
3.	Vernier Caliper	Cl. 7
4.	Micrometer	
5.	Steel ruler	
6.	Magnetic Stand with dial indicator for Shaft Extension Runout, Concentricity, Perpendicularity.	
7.	Insulation Tester	Cl. 16.1 a, 16.2 a
8.	High voltage tester	Cl. 16.1 b, 16.2 b
9.	Auto transformer	Cl. 16.2 c, 16.2 d, 16.1 f, 16.1 g, 16.1 h, 16.1 j, 16.1 k.
10.	Current transformer	Cl. 16.2 c, 16.2 d, 16.1 f, 16.1 g, 16.1 h, 16.1 j, 16.1 k.
11.	Frequency Meter	Cl. 16.2 c, 16.2 d, 16.1 f, 16.1 g, 16.1 h, 16.1 j, 16.1 k, 18.1
12.	Voltmeter	Cl. 16.2 c, 16.2 d, 16.1 f, 16.1 g, 16.1 h, 16.1 j, 16.1 k, 18.1
13.	Ammeter	Cl. 16.2 c, 16.2 d, 16.1 f, 16.1 g, 16.1 h, 16.1 j, 16.1 k, 18.1
14.	Wattmeter	Cl. 16.2 c, 16.2 d, 16.1 d, 16.1 f, 16.1 g, 16.1 h., 18.1
15.	Tacho meter	Cl. 16.1 d, 16.1 g, 16.2 c
16.	Digital Slip speed Meter with Slip coil	Cl. 16.2 c, 16.1 g, 16.1 h
17.	Milli Ohm meter	Cl. 16.1 c, 16.1 j, 16.1 k.
18.	Thermo meter	Cl. 16.1 c, 16.1 j, 16.1 k
19.	Motor load testing bed & Spring balance/ Torque bench brake drum	Cl. 16.1 g, 16.1 h, 16.2 d, 18.1
20.	Sump	Cl. 16.1 & 16.2
21.	Milli Ammeter	Cl. 22

Note: The above list is indicative only and may not be treated as exhaustive.

ANNEX – C

निरीक्षण और परीक्षण की स्कीम
Scheme Of Inspection And Testing

1. QUALITY ASSURANCE PLAN

- 1.1 It is expected that manufacturers (licensees/applicants) will implement a Quality Assurance Plan i.e. a plan of regular testing and in-process controls, designed to ensure that the product bearing the Standard Mark conforms to all requirements of the Indian Standard.
- 1.2 The manufacturers shall define a Quality Assurance Plan defining the control unit (i.e. lot/batch etc.) and the levels of control (i.e. the frequency and number of samples for conducting the different tests as per the Indian Standard) and submit the same to BIS Branch Office for information. The manufacturer shall comply with the same and maintain test records in accordance with para 2.4.

1.3 RECOMMENDED LEVELS OF CONTROL/CONTROL UNIT:

- 1.3.1 For the guidance of manufacturers, the recommended definition of control unit is: **All Motors of same model manufactured in a day, shall constitute a control unit.** The manufacturer has the choice to define their own control unit/batch/lot and submit the same to the BIS.
- 1.3.2 For the guidance of manufacturers in preparing the Quality Assurance Plan, recommended levels of control are given in Table 1.
- 1.3.3 The manufacturer shall ensure inspection and testing as per the Quality Assurance Plan submitted by them on the whole production of the factory which is covered by this plan. Alternatively, the manufacturer has the option of adherence to the quality plan as per levels of control recommended in column 3 of Table 1.
- 1.4 However, all manufacturers shall ensure compliance of their products to all the requirements of the Indian Standard.

2. ENSURING COMPLIANCE THROUGH TESTING- It is expected that manufacturers (licensees/applicants) will establish a suitably equipped and staffed in house laboratory (In house testing facility) for testing at least those parameters of the Indian Standard which require routine testing for ensuring quality of the product. This includes in-process controls as may be defined and put in place by the manufacturer and testing parameters/requirements which can only be performed in the factory.

2.1 For the guidance of manufacturers, Table 1 giving the recommended levels of control is given below. Column 2 of Table 1 indicates routine tests where test equipment is required in house as “R” or other tests which can be subcontracted as “S”. Subcontracting is permitted to BIS recognized/empaneled laboratory or any other laboratory having valid NABL accreditation as per IS/ISO/IEC 17025.

2.2 For MSME manufacturers, the requirement of maintaining a laboratory/in-house testing facility for routine tests (indicated as “R” in Column 2 of Table 1) is also optional.

2.2.1 MSME manufacturers may utilize common cluster-based facilities as per guidelines for the utilization of cluster-based test facilities by MSMEs or the provisions of Sharing of testing facilities or get testing done from BIS recognized/empaneled laboratory or any other laboratory having valid NABL accreditation as per IS/ISO/IEC 17025.

2.3 Large Scale manufacturers shall maintain an in-house laboratory equipped at least with test facilities for routine tests (indicated as “R” in Column 2 of Table 1), where different tests given in the specification shall be carried out in accordance with the method given in the specification. They shall also implement a calibration plan for the in-house test equipment.

2.3.1 Alternatively, in lieu of an in-house laboratory, large scale manufacturers can also utilize the provisions of Sharing of testing facilities as per the Guidelines for Grant of Licence available on BIS website www.bis.gov.in. (Under Conformity Assessment>Product Certification Process). Even for subcontracted tests, provisions for sharing of testing facilities can be utilized.

2.4 TEST RECORDS- The manufacturers maintaining an in-house laboratory or utilizing common cluster-based facilities or shared test facilities shall maintain test records for the tests carried out to establish conformity. For the tests being subcontracted to BIS recognized/empaneled laboratory or any other laboratory having valid NABL accreditation as per IS/ISO/IEC 17025, test reports issued by the laboratories shall be available for inspection by BIS.

3. MARKING –

a) The Standard Mark as given in the Schedule of the licence shall be incorporated legibly and indelibly on **each Motors and also on its packaging** provided always that the material so marked conforms to each requirement of the specification.

b) Marking shall also be done as per Cl. 15 of IS 9283:2024.

3.1 Additional Marking requirements: The product shall also be marked with the following additional requirement on packaging:

a) “For BIS certification details please visit www.bis.gov.in”

4. REJECTION - All the production which conforms to the Indian Standard and covered under the scope of this licence shall be marked with the Standard Mark. Disposal of non-conforming product shall be done in such a way so as to ensure that there is no violation of provisions of BIS Act, 2016.

TABLE 1

(1)				(2)	(3)		
Test Details				Test equipment requirement R: required (or) S: Sub-contracting permitted	Levels of Control		
Cl.	Requirement	Test Methods			No. of Sample	Frequency	Remarks
		Clause	Reference				
5.1.2	Material	Cl. 5.1.2 Table 1	IS 9283	S	No further testing is required, if accompanied with Test Certificate or ISI Marked.		
5.2	Earthing	5.2	IS 9283	R	Manufacturer shall exercise adequate levels of control to ensure that entire production should conform to the Standard. Appropriate records of testing indicating conformity shall be maintained.		
5.3	Foreign Matter	5.3	IS 9283	R			
5.4	Construction a) Type of cable b) Size of conductor of cable c) Length of cable d) Joint in Cable	5.4 5.4.1 5.4.1 5.4.2 5.4.3	IS 9283	R			
5.5	Finish of Bearings	5.5	IS 9283& IS 3073	R			
5.6	Balancing of Rotor	5.6	IS/ISO 21940-11	R			
7	Dimensions and Tolerances						
7.1	Dimensions	7.1	IS 9283	R			
7.2	Tolerances			R			
	a) For motors with anti-friction bearing	7.2.1, Table 6,7	IS 9283, IS 2223				
	b) For motors with journal bearing	7.2.2	IS 9283				
13	Terminal Marking	13	IS 9283	R			
12	Efficiency	12.1	IS 9283	R			
16.2(a)	Insulation resistance before HV test	21	IS 9283	R			
16.2(b)	High Voltage Test	20	IS 9283	R			
16.2(c)	No-load running of Motor and reading of Current, Voltage, Power & Speed	8.1	IS 4029	R			
		8.1	IS 7572				

(1)				(2)	(3)		
Test Details				Test equipment requirement R: required (or) S: Sub-contracting permitted	Levels of Control		
Cl.	Requirement	Test Methods			No. of Sample	Frequency	Remarks
		Clause	Reference				
16.2(d)	Locked rotor reading of voltage, current and power input	8.3	IS 4029	R	Each Motor	—	
		8.2	IS 7572				
16.2(e)	Reduced voltage running up test at no-load	16.2(e)	IS 9283	R			
16.1(a)	Insulation resistance Test after HV Test	21	IS 9283	R	5 Motor	Each Control unit	
16.1(c)	Measurement of Stator Resistance	7	IS 4029	R			
		7	IS 7572				
16.1(f)	Locked rotor torque	8.3	IS 4029	R			
		8.2	IS 7572				
16.1(g)	Full Load Reading of voltage, current, power input, speed and frequency	8.8	IS 4029	R			
		8.4	IS 7572				
16.1(h)	Performance characteristics	11.1	IS 9283	R			
16.1(j)	Temperature rise test at rated voltage	19	IS 9283	R			
16.1(k)	Temperature rise test at reduced voltage	19	IS 9283	R			
16.1(m)	Momentary Overload Test	18	IS 9283	R			
16.1(n)	Leakage Current Test at rated voltage at no load	22	IS 9283	R			